



Science

Stage 9

Paper 1

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

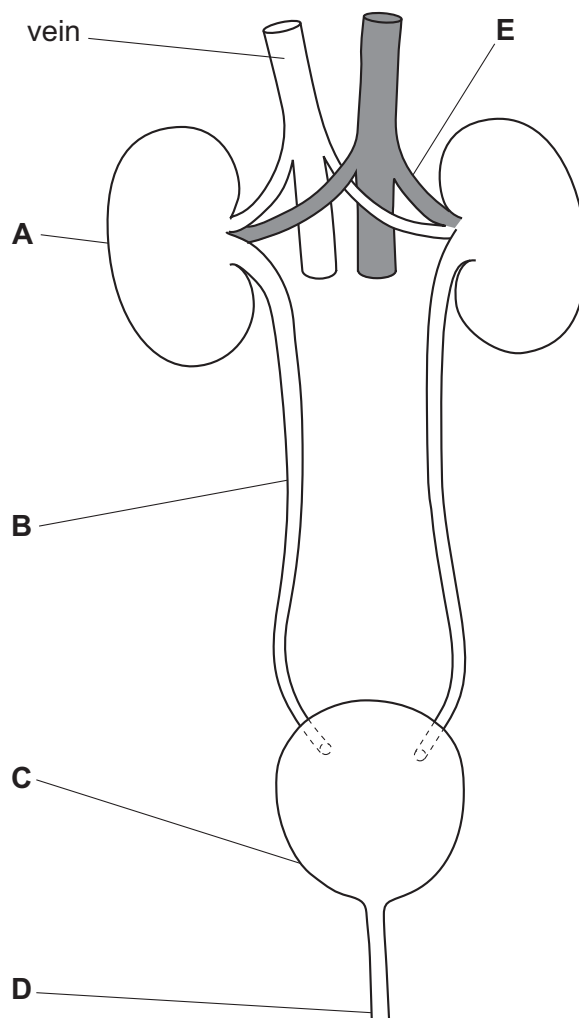
INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

1 Look at the diagram of the human excretory (renal) system.



(a) Complete the table to name the parts labelled **A**, **B**, **C**, **D** and **E**.

part	name of part
A	
B	
C	
D	
E	

[3]

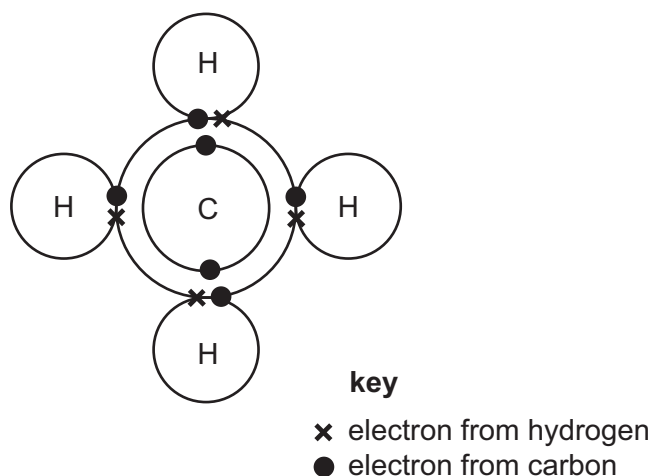
(b) Urea ammonium nitrate is an organic fertiliser.

What do plants make using nitrates?

..... [1]

2 Methane has a simple structure.

Look at the model of a molecule of methane.



(a) How many electrons are in a molecule of methane?

..... [1]

(b) Write down the type of bonding in a molecule of methane.

Explain your answer.

type of bonding

explanation

..... [2]

(c) Methane has a simple structure.

Predict **one physical** property of methane.

..... [1]

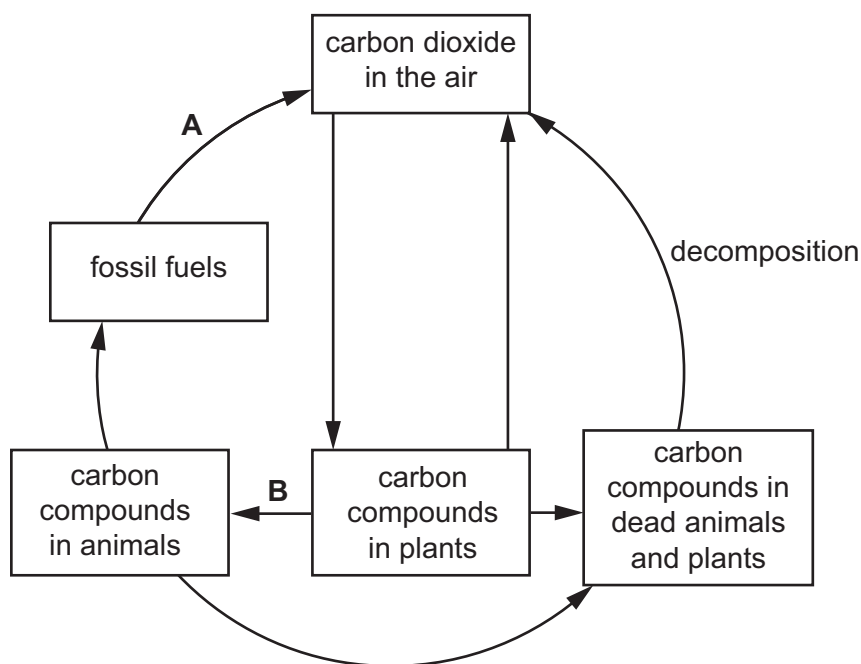
- 3 Jamila spills some liquid on the back of her hand.

After a short time, Jamila's hand feels cold.

Name the process that makes her hand feel cold.

..... [1]

- 4 Look at the diagram of part of the carbon cycle.



- (a) One of the arrows in the diagram shows photosynthesis.

Write the letter **P** on the arrow that shows photosynthesis.

[1]

- (b) Name process **A** and process **B**.

process **A**

process **B**

[2]

5 Lily reacts magnesium with copper sulfate solution.

(a) Name this type of reaction.

..... [1]

(b) Name the **two** products of this reaction.

..... and [2]

(c) Lily adds zinc to magnesium sulfate solution.

There is **no** reaction.

Explain why there is **no** reaction.

.....
..... [1]

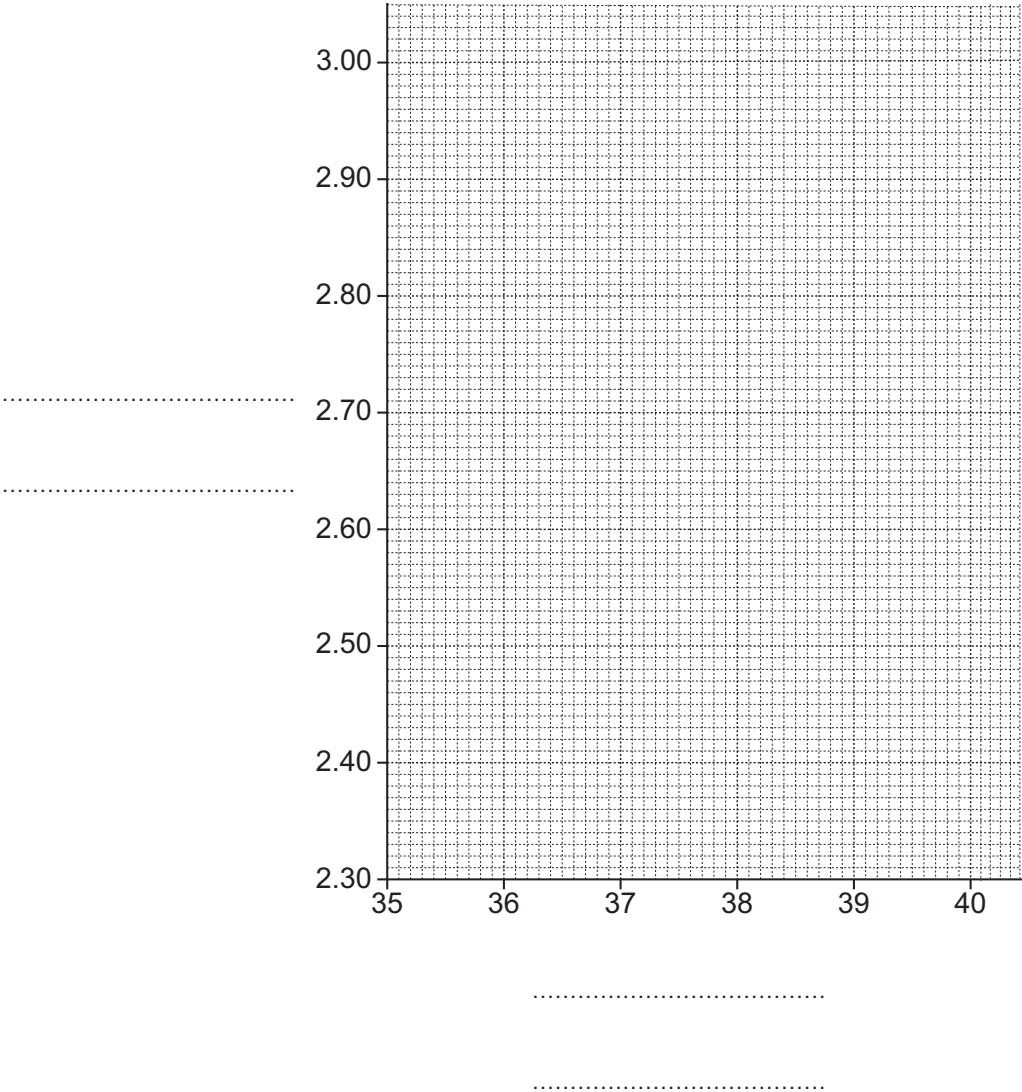
6 Fetal development is affected by the health of the mother.

The table shows information about the mass of a fetus during pregnancy for a mother who smokes.

age of the fetus in weeks	mass of the fetus in kg
36	2.40
37	2.55
38	2.60
39	2.85
40	3.00

(a) Draw the graph of this information:

- label the x-axis
- label the y-axis
- plot all the points from the table.



[2]

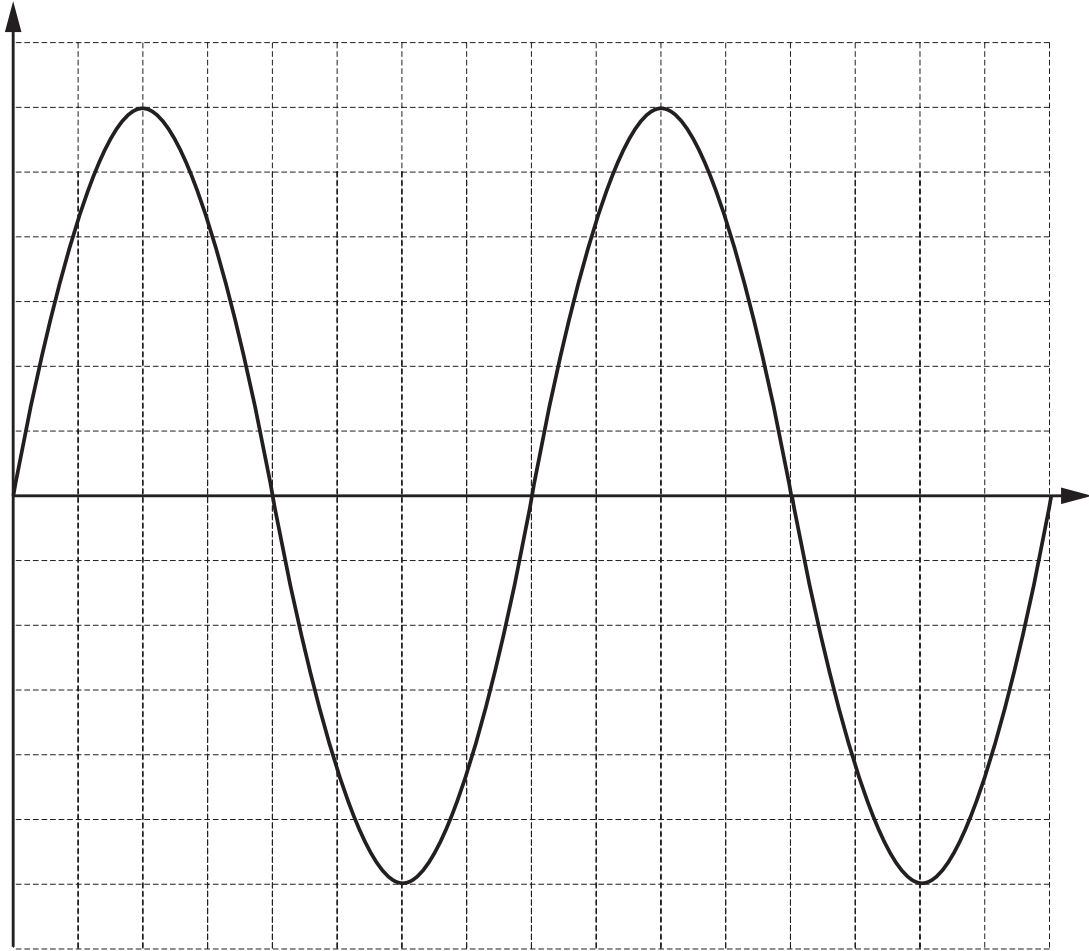
(b) There is **one** anomalous result.

Circle the anomalous result on the graph. [1]

(c) Draw a line of best fit on the graph. [1]

7 This question is about different waveforms and sounds.

(a) Look at the diagram of a waveform.



Draw a double-headed arrow ($\longleftrightarrow$ or $\updownarrow$) on the diagram to show the amplitude of the waveform.

[1]

(b) Aiko hears a high pitch sound and then a low pitch sound.

Write a sentence to compare the frequencies of the two sounds.

.....
 [1]

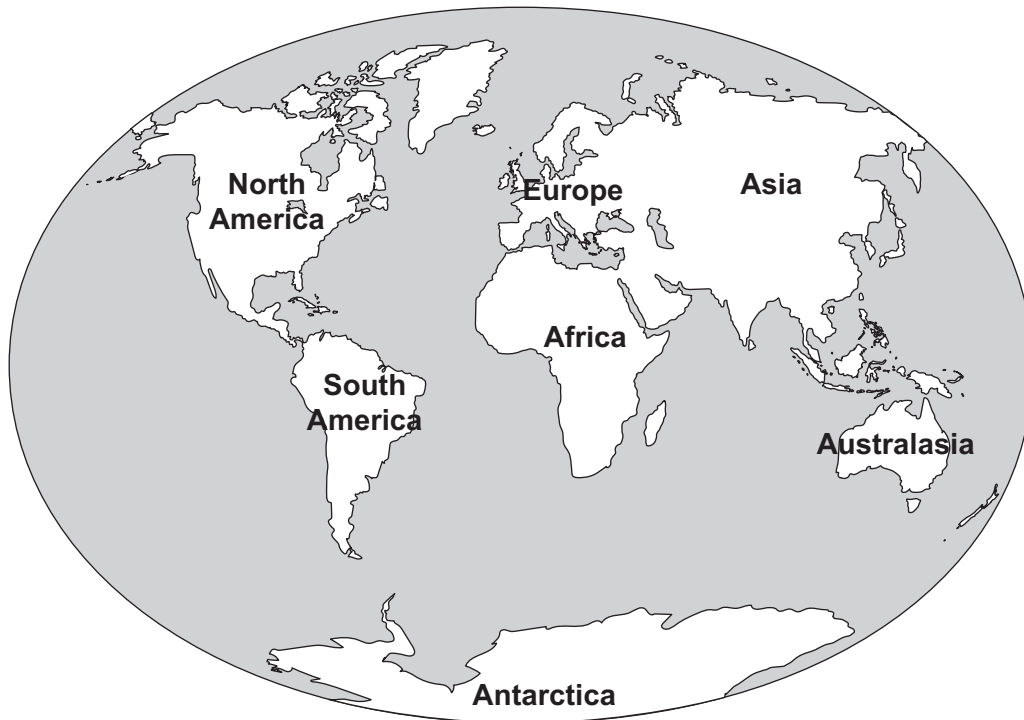
(c) Aiko plays a sound on her phone.

Complete the sentence.

The louder the sound the larger the

[1]

8 The map shows the Earth's continents.



Fossils of a plant species that grew 200 million years ago have been found in South America, Africa, Antarctica and Australasia.

Explain why fossils of this plant species are found on these four continents.

.....

.....

.....

..... [2]

9 Water travels through plants.

Look at the list of parts of a plant.

- A** leaf xylem
- B** stem xylem
- C** stoma on the surface of leaf
- D** root hair cell
- E** root xylem

Write down parts **A** to **E** in the correct order to show the pathway of water through a plant.

One has been done for you.

.....		B		
-------	-------	----------	-------	-------

[2]

10 The table shows the melting points and boiling points of some Group 1 elements.

element	melting point in °C	boiling point in °C
lithium	181	1347
sodium		881
potassium	63	766
rubidium	39	688

(a) Describe the trend in the boiling point from the top to the bottom of Group 1.

..... [1]

(b) Predict the melting point of sodium.

..... °C [1]

11 Yuri investigates current in a parallel electrical circuit.

(a) Draw his circuit with:

- a battery of cells
- a closed switch
- two lamps in parallel with each other
- a meter to measure the current through one of the lamps.

Use standard electrical symbols.

[3]

(b) Yuri uses the same components to make a series electrical circuit with two lamps.

Write a sentence to compare the brightness of the two lamps in the two circuits.

.....
..... [1]

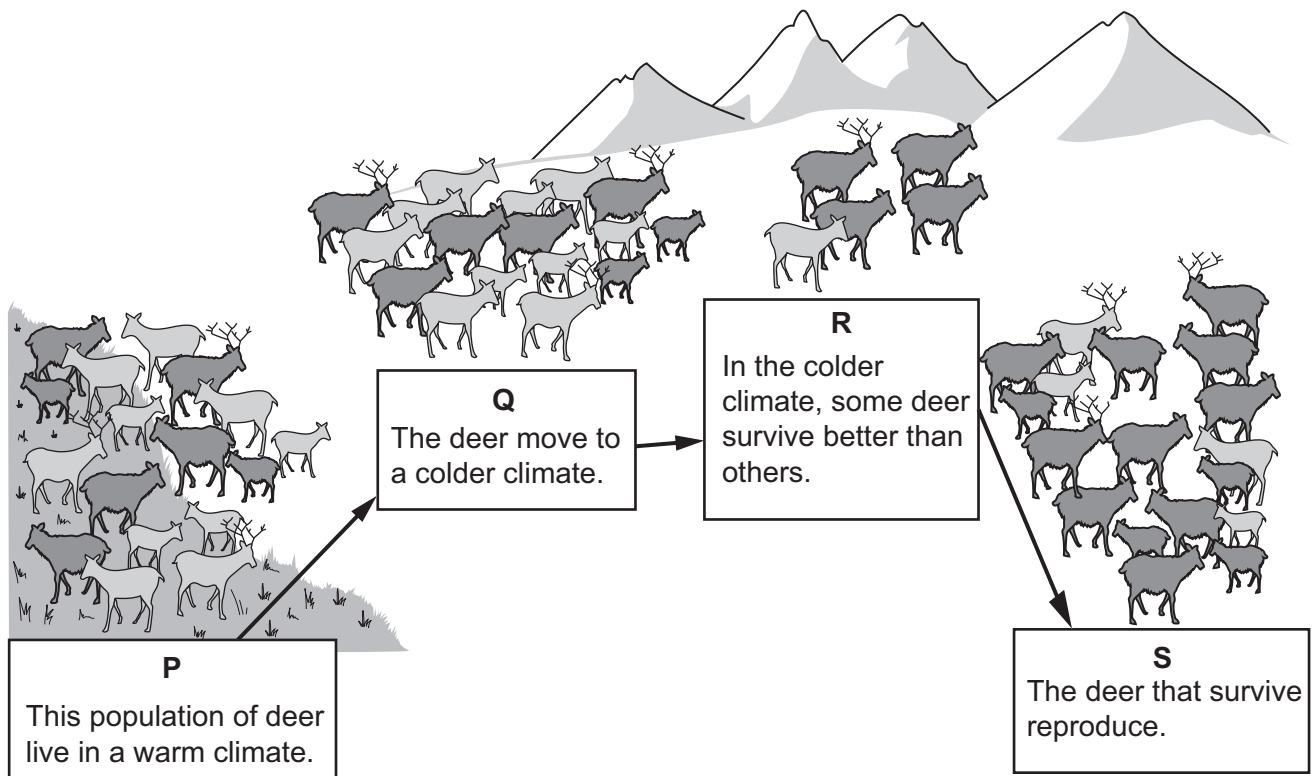
12 Complete the sentences about nebulae.

Nebulae are enormous clouds of and

Some nebulae act as stellar nurseries for new

[3]

13 The diagram shows how the appearance of a deer population changes over time.



(a) Look at the diagram.

The deer show variation.

Describe **one** example of variation shown in the diagram.

..... [1]

- (b) The appearance of the deer population changes over time from **P** to **S**.

The theory of natural selection explains this change in appearance.

Match each **statement about natural selection** to the correct **letter** in the diagram.

You may use each letter once, more than once or not at all.

statement about natural selection	letter
Some adaptations give advantages in one environment and not in another.	
Genes controlling features which help survival pass into offspring.	
Individuals with adaptations less advantageous to the environment die before they reproduce.	

[2]

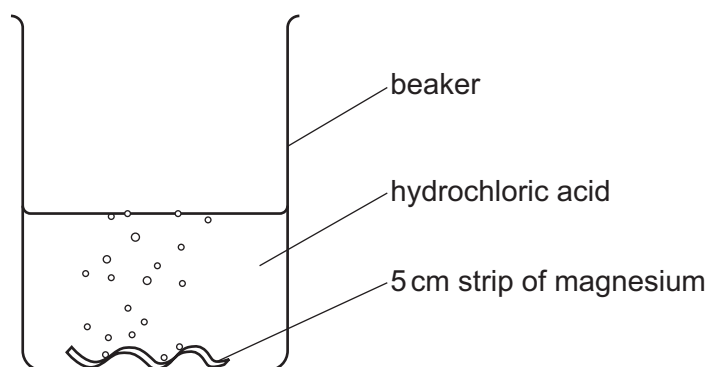
14 Oliver investigates the effect of temperature on the rate of reaction.

In his first experiment, Oliver:

- puts 20 cm^3 of dilute hydrochloric acid into a beaker
- measures the temperature of the acid
- places a 5 cm strip of magnesium into the acid
- measures the reaction time.

Oliver repeats the experiment four more times.

Each time he uses acid at a different temperature.



Look at his table of results.

temperature of acid in $^{\circ}\text{C}$	reaction time in s
20	120
25	75
30	63
35	35
40	29

(a) Describe the relationship between the temperature of the acid and the reaction time.

.....

.....

..... [1]

- (b) Oliver repeats the investigation using **more concentrated** hydrochloric acid.

Write a prediction for this investigation.

Explain your prediction using the particle model.

prediction

.....

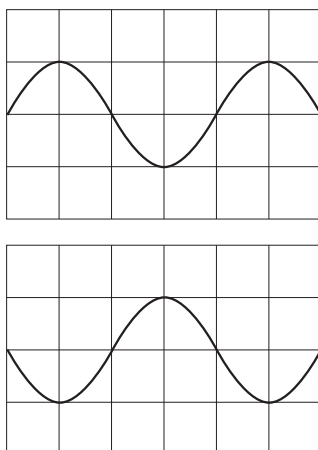
explanation

.....

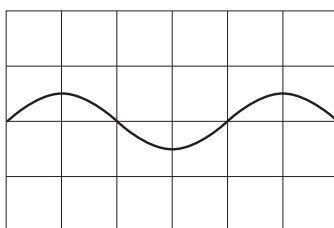
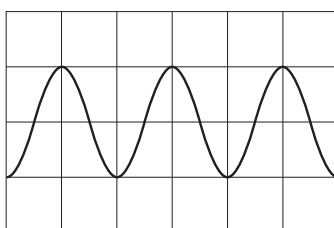
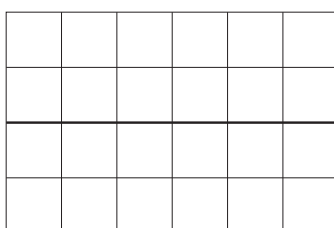
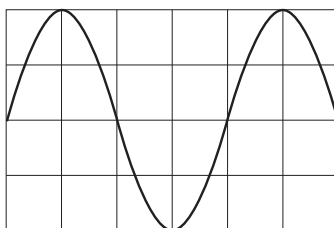
.....

[3]

15 Look at the diagram of two waveforms.



Tick (✓) the box which shows what happens when these two waveforms interact.


☐

☐

☐

☐

[1]

16 This question is about the impact of climate change.

Write down **two** impacts of climate change.

1

2

[2]

17 This question is about floating and sinking.

(a) Look at the table of densities for different materials.

material	density in kg/m^3
concrete	2400
iron	7000
plastic	920
water	1000
wood	600

Which materials from the table float in water?

Explain your answer.

materials

explanation

.....

.....

[2]

(b) An object sinks in water but floats in sea water.

Predict a value for the density of sea water.

Circle the correct answer.

100 kg/m^3

900 kg/m^3

1000 kg/m^3

1030 kg/m^3

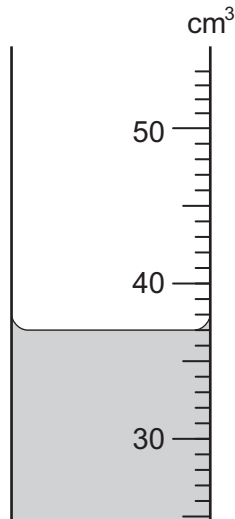
[1]

(c) An object is put into a container full of water.

The object floats and a volume of water comes out of the container.

The water is collected in a 100 cm³ measuring cylinder.

Look at the diagram of part of the measuring cylinder.



What volume of water is collected?

..... cm³ [1]

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The Periodic Table of Elements

Group

1	2	3					4	5	6	7	8																								
		1 H hydrogen 1																																	
		Key atomic number atomic symbol name relative atomic mass																																	
3	Li lithium 7	4	Be beryllium 9									5	B boron 11	6	C carbon 12	7	N nitrogen 14	8	O oxygen 16	9	F fluorine 19	10	Ne neon 20												
11	Na sodium 23	12	Mg magnesium 24									13	Al aluminium 27	14	Si silicon 28	15	P phosphorus 31	16	S sulfur 32	17	Cl chlorine 35.5	18	Ar argon 40												
19	K potassium 39	20	Ca calcium 40	21	Sc scandium 45	22	Ti titanium 48	23	V vanadium 51	24	Cr chromium 52	25	Mn manganese 55	26	Fe iron 56	27	Co cobalt 59	28	Ni nickel 59	29	Cu copper 64	30	Zn zinc 65	31	Ga gallium 70	32	Ge germanium 73	33	As arsenic 75	34	Se selenium 79	35	Br bromine 80	36	Kr krypton 84
37	Rb rubidium 85	38	Sr strontium 88	39	Y yttrium 89	40	Zr zirconium 91	41	Nb niobium 93	42	Mo molybdenum 96	43	Tc technetium —	44	Ru ruthenium 101	45	Rh rhodium 103	46	Pd palladium 106	47	Ag silver 108	48	Cd cadmium 112	49	In indium 115	50	Sn tin 119	51	Sb antimony 122	52	Te tellurium 128	53	I iodine 127	54	Xe xenon 131
55	Cs caesium 133	56	Ba barium 137	57–71	lanthanoids	72	Hf hafnium 178	73	Ta tantalum 181	74	W tungsten 184	75	Re rhenium 186	76	Os osmium 190	77	Ir iridium 192	78	Pt platinum 195	79	Au gold 197	80	Hg mercury 201	81	Tl thallium 204	82	Pb lead 207	83	Bi bismuth 209	84	Po polonium —	85	At astatine —	86	Rn radon —
87	Fr francium	88	Ra radium	89–103	actinoids	104	Rf rutherfordium	105	Db dubnium	106	Sg seaborgium	107	Bh bohrium	108	Hs hassium	109	Mt meitnerium	110	Ds darmstadtium	111	Rg roentgenium	112	Cn copernicium	113	Nh nihonium	114	Fl flerovium	115	Mc moscovium	116	Lv livermorium	117	Ts tennessine	118	Og oganesson
				lanthanoids																															
				57	La lanthanum 139	58	Ce cerium 140	59	Pr praseodymium 141	60	Nd neodymium 144	61	Pm promethium —	62	Sm samarium 150	63	Eu europium 152	64	Gd gadolinium 157	65	Tb terbium 159	66	Dy dysprosium 163	67	Ho holmium 165	68	Er erbium 167	69	Tm thulium 169	70	Yb ytterbium 173	71	Lu lutetium 175		
				89	Ac actinium	90	Th thorium 232	91	Pa protactinium 231	92	U uranium 238	93	Np neptunium —	94	Pu plutonium	95	Am americum	96	Cm curium	97	Bk berkelium	98	Cf californium	99	Es einsteinium	100	Fm fermium	101	Md mendelevium	102	No nobelium	103	Lr lawrencium		
				actinoids																															



Science

Stage 9

Paper 1

2024

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	(A) kidney (B) ureter (C) bladder (D) urethra (E) (renal) artery	3	all five correct = 3 marks three or four correct = 2 marks one or two correct = 1 mark
1(b)	protein(s)	1	

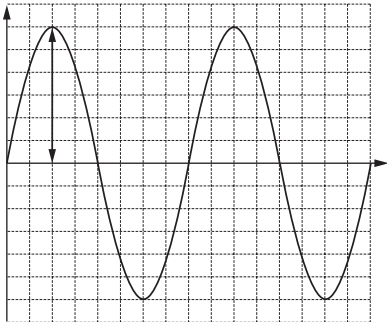
Question	Answer	Marks	Further Information
2(a)	10	1	
2(b)	(type of bonding) covalent (bonding) (explanation) a pair of electrons shared (by two atoms)	2	each correct answer = 1 mark Accept two electrons that are shared
2(c)	any one from low melting point low boiling point gas at room temperature	1	 Accept poor conductor of electricity or does not conduct electricity or can flow easily or has a low density

Question	Answer	Marks	Further Information
3	evaporation	1	Accept endothermic Ignore perspiration

Question	Answer	Marks	Further Information
4(a)	label the arrow from carbon dioxide in the air to carbon compounds in plants with the letter P	1	
4(b)	(process A) combustion (process B) feeding	2	each correct answer = 1 mark Accept burning Accept eating Ignore nutrition or taking in nutrients

Question	Answer	Marks	Further Information
5(a)	displacement	1	Ignore chemical
5(b)	copper magnesium sulfate	2	each correct answer = 1 mark Note answers can be given in either order
5(c)	zinc is less reactive than magnesium	1	Accept ora Accept zinc is lower down the reactivity series than magnesium

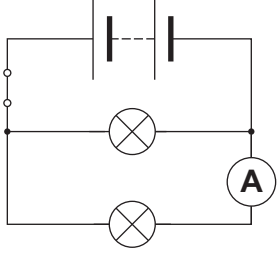
Question	Answer	Marks	Further Information
6(a)	mass of fetus in kg age of fetus in weeks	2	labelling of x-axis and labelling of y-axis = 1 mark all points correctly plotted $\pm \frac{1}{2}$ small square = 1 mark Accept plots with small x, dot, or dot with circle
6(b)	point at (38, 2.6) circled	1	Accept any indication of anomalous plot, e.g. circle on table of results, but answer on graph takes precedence Accept ecf on plotting from part (a)
6(c)		1	Accept ecf on plotting from part (a) Accept ecf on plotting from part (b) Do not accept dot to dot Note line of best fit judged by eye and does not consider the anomalous result

Question	Answer	Marks	Further Information
7(a)	 arrow drawn vertically from x-axis to the crest of waveform or arrow drawn vertically from x-axis to the trough of waveform	1	
7(b)	high pitch sound has higher frequency (than low pitch sound)	1	Accept ora Accept the frequency decreases
7(c)	amplitude	1	

Question	Answer	Marks	Further Information
8	(idea that) these continents were once joined together (idea that) continents moved apart due to movement of plates or moved apart due to plate tectonics	2	each correct answer = 1 mark Accept (idea of) continental drift

Question	Answer	Marks	Further Information
9	D E (B) A C	2	DE in correct order = 1 mark AC in correct order = 1 mark

Question	Answer	Marks	Further Information
10(a)	decreases	1	
10(b)	98	1	Accept answers in the range 80–120

Question	Answer	Marks	Further Information
11(a)		3	any complete circuit drawn with at least one component and no gaps = 1 mark correct symbol for a battery of cells and closed switch = 1 mark Do not accept open switch correct symbol for lamps in parallel and correct symbol for ammeter in series with one lamp = 1 mark Ignore junction of conductor symbols
11(b)	in a series circuit the lamps are less bright	1	Accept ora Note answer must be comparative
Question	Answer	Marks	Further Information
12	dust and gas stars	3	each correct answer = 1 mark Accept dust and gas in either order

Question	Answer	Marks	Further Information								
13(a)	height or size or mass or fur colour	1	Accept fur thickness or antlers Ignore age								
13(b)	<table><tr><td>statement about natural selection</td><td>letter</td></tr><tr><td>Some adaptations are advantageous in one environment and not in another.</td><td>R</td></tr><tr><td>Genes controlling features which help survival pass into offspring.</td><td>S</td></tr><tr><td>Individuals with adaptations less advantageous to the environment die before they reproduce.</td><td>R</td></tr></table>	statement about natural selection	letter	Some adaptations are advantageous in one environment and not in another.	R	Genes controlling features which help survival pass into offspring.	S	Individuals with adaptations less advantageous to the environment die before they reproduce.	R	2	all three correct = 2 marks one or two correct = 1 mark
statement about natural selection	letter										
Some adaptations are advantageous in one environment and not in another.	R										
Genes controlling features which help survival pass into offspring.	S										
Individuals with adaptations less advantageous to the environment die before they reproduce.	R										

Question	Answer	Marks	Further Information
14(a)	as the temperature increases the reaction time decreases	1	Accept ora
14(b)	(prediction) as the concentration increases the reaction time decreases or as the concentration increases the rate increases (explanation) (as concentration increases) there are more particles per unit volume or particles are more crowded or particles are closer together more collisions	3	each correct answer = 1 mark Accept more particles in the same volume Accept more frequent collisions or more chance of collision or collisions more often or more collisions per second = 2 marks Do not accept particles have more energy or particles are moving faster Ignore references to successful collisions

Question	Answer	Marks	Further Information
15	3 rd box ticked	1	more than one box ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence

Question	Answer	Marks	Further Information
16	any two from sea levels rise or sea levels change flooding drought extreme weather events	2	each correct answer = 1 mark Accept other suitable impacts of climate change
Question	Answer	Marks	Further Information
17(a)	(materials) plastic and wood (explanation) less dense than water	2	each correct answer = 1 mark both materials needed either order for the mark Note explanation marking point is independent of materials
17(b)	1030 kg/m ³	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
17(c)	37(.0) (cm ³)	1	

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Science

Stage 9

Paper 2

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

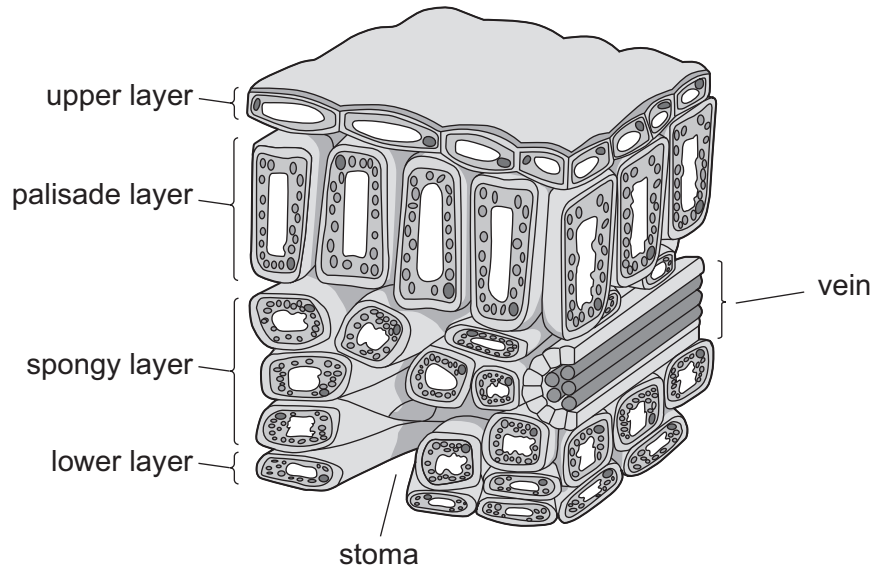
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- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Look at the diagram of part of the cross-section of a plant leaf.



- (a) The palisade layer contains many chloroplasts.

- (i) What process happens inside chloroplasts?

..... [1]

- (ii) Name **two** gases exchanged through the stoma.

1

2

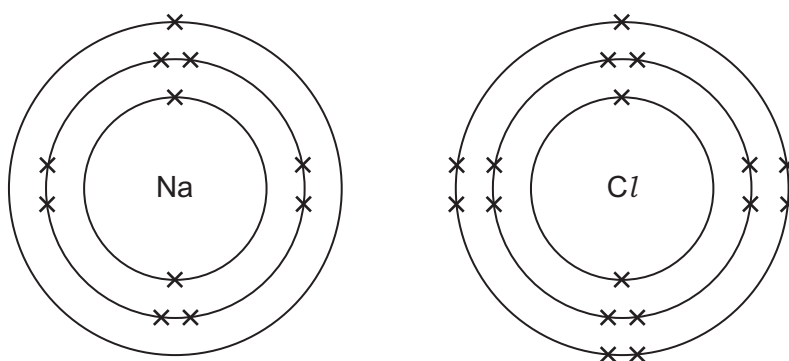
[2]

- (b) Chloroplasts contain chlorophyll.

What is the function of chlorophyll?

..... [1]

- 2 The diagram shows a model of the atomic structure of a sodium atom and a chlorine atom.



A sodium atom and a chlorine atom react together to make a sodium ion, Na^+ , and a chloride ion, Cl^- .

- (a) Describe how a sodium atom and a chlorine atom become charged ions.

sodium atom

.....

chlorine atom

.....

[2]

- (b) Describe how ions are held together in an ionic compound.

.....

.....

[2]

3 This question is about thermal transfer.

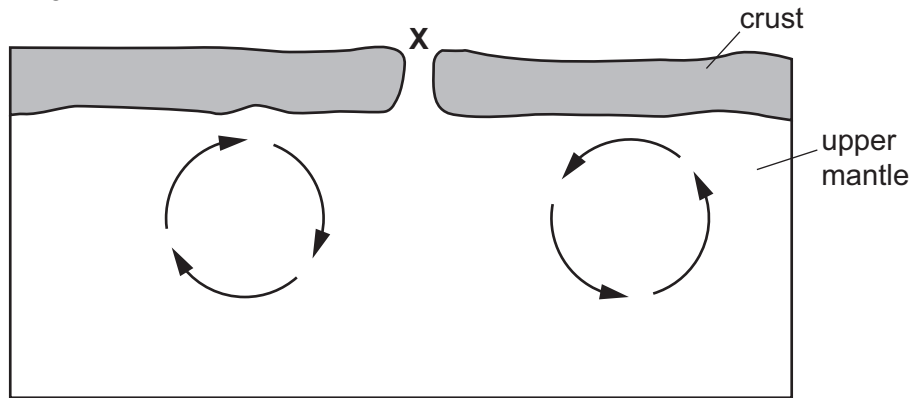
Decide if each sentence is **true** or **false**.

Tick (✓) the correct box.

sentence	true	false
Heat is the word used to measure how hot an object is.	☐	☐
Thermal energy is measured in °C.	☐	☐
When thermal energy is transferred, more thermal energy is created.	☐	☐
Thermal energy always transfers from hotter objects to colder objects.	☐	☐
Heat dissipation happens when thermal energy transfers from a hotter region to a colder region.	☐	☐
A liquid is a good thermal energy conductor.	☐	☐
Convection happens in liquids and gases only.	☐	☐
Radiation needs particles to transfer thermal energy.	☐	☐

[4]

- 4 Look at the diagram of a model of the Earth's crust and upper mantle.



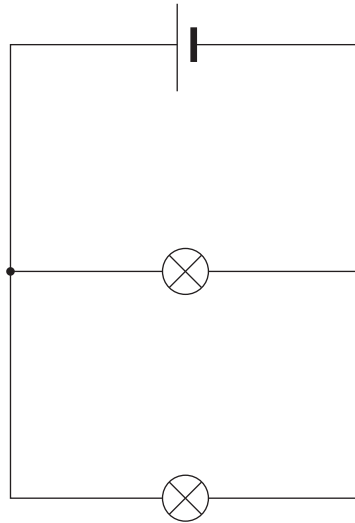
- (a) Name **one** event that happens at **X**.

..... [1]

- (b) What do the arrows on the diagram represent?

..... [1]

- 5 The diagram shows a circuit with two lamps.



Complete the circuit diagram to show how to measure the voltage across one of the lamps. [2]

6 Safia is making a salt called copper chloride.

She adds insoluble copper carbonate to dilute hydrochloric acid.

Water and carbon dioxide are also produced.

(a) Write the word equation for this reaction.

..... [2]

(b) Complete the information about the method Safia uses to make copper chloride.

Safia:

- adds excess copper carbonate to dilute hydrochloric acid
- separates the excess copper carbonate from the reaction mixture by

.....

- pours the filtrate into an evaporating basin
- obtains a saturated solution from the filtrate by
- leaves the saturated solution in a warm place to

[3]

(c) Suggest **one** piece of safety equipment Safia uses when making copper chloride.

..... [1]

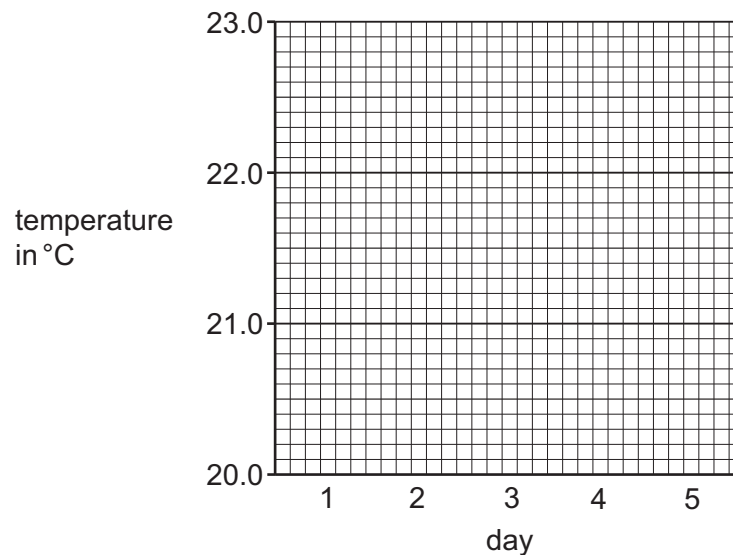
- 7 Mike investigates the temperature of a classroom every day for five days.

Look at the table of his results.

day	temperature in °C
1	21.0
2	21.5
3	22.3
4	21.8
5	22.4

- (a) Mike starts to plot the results in a bar chart.

Complete his bar chart.



[2]

- (b) Carlos tells Mike to measure the temperature twice each day at the same time of day.

Explain why this improves the investigation.

.....
 [1]

- 8 Scientists think the Moon was formed during a collision between the Earth and another small planet.

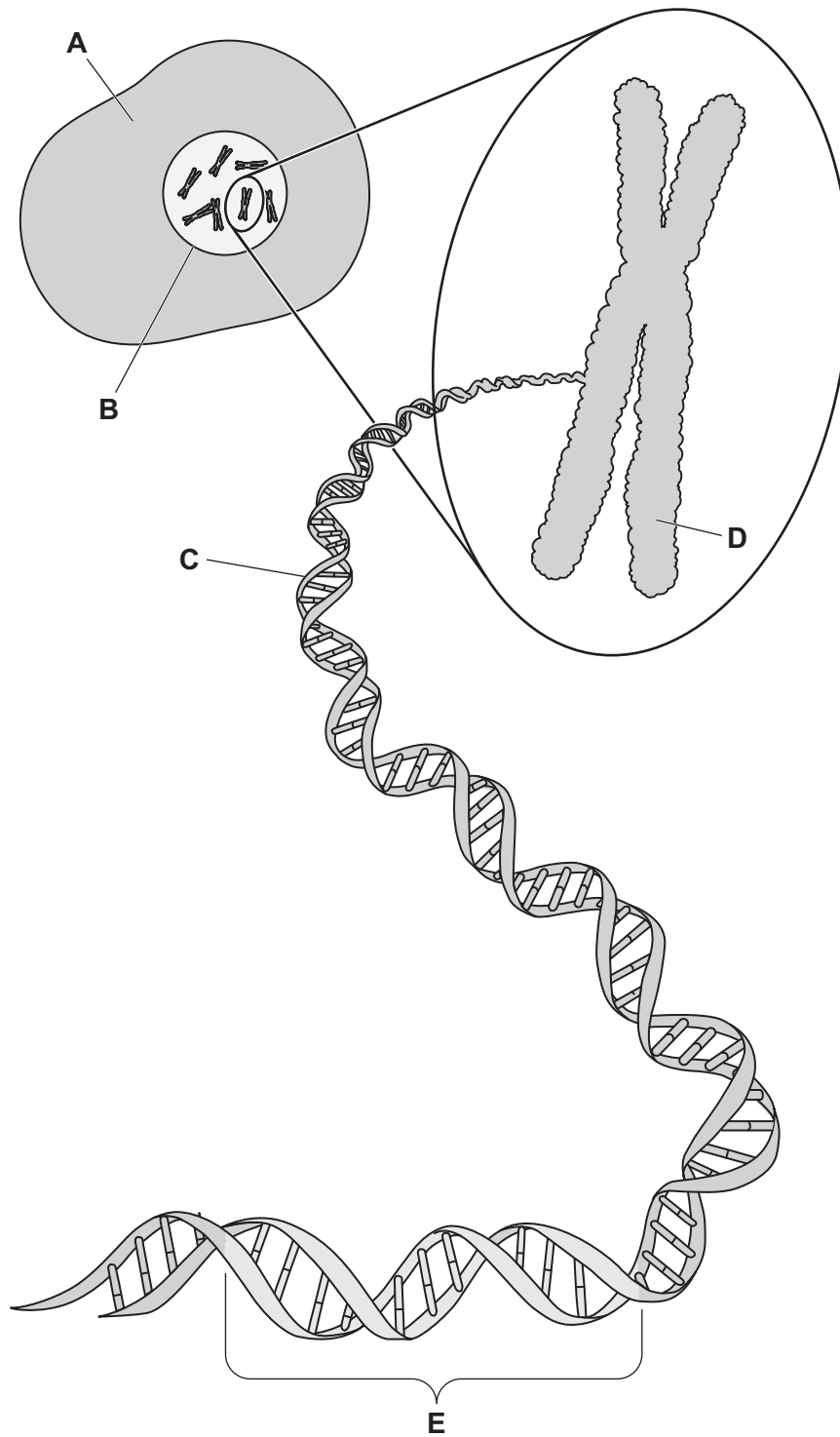
This is called the collision theory for the formation of the Moon.

Describe **three** reasons why rocks collected from the Moon support this collision theory.

- 1
- 2
- 3
-

[3]

- 9 The diagram shows the genetic material inside an animal cell.



NOT TO SCALE

Look at parts **A**, **B**, **C**, **D** and **E**.

(a) Circle the letter that shows a chromosome.

A**B****C****D****E**

[1]

(b) Circle the letter that controls a single characteristic such as eye colour.

A**B****C****D****E**

[1]

(c) Write down the name of the chemical that is in a chromosome.

..... [1]

- 10** Ahmed wants to find the density of dry sand.

The dry sand has a mass of 150 g and a volume of 100 cm³.

Calculate the density of the dry sand.

Include the unit of density in your answer.

density = unit
[3]

11 Mia investigates an electrical circuit.

(a) She makes this prediction,

**‘Adding more lamps into a series circuit
does not change the current in the circuit.’**

Tick (✓) to show if Mia's prediction is correct.

yes

☐

no

☐

Give a reason for your answer.

.....

.....

[1]

(b) The current flowing through a lamp is 4.5 A.

The voltage across the lamp is 1.5 V.

Calculate the resistance of the lamp.

resistance = Ω [2]

- 12 (a) A scientist develops a model about climate change.

The scientist makes this prediction in 2013.

**‘By the year 2100 the average temperature
at the Earth’s surface will rise by 1.5 °C.’**

The same scientist changes the prediction in 2023.

**‘By the year 2100 the average temperature
at the Earth’s surface will rise by 2.5 °C.’**

Suggest **two** reasons why the scientist changes the prediction.

1

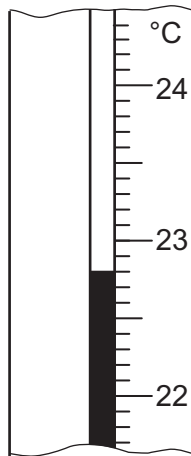
2

[2]

- (b) The scientist plans an investigation to monitor air temperature during a period of one year.

- (i) The scientist uses a thermometer to measure air temperature.

Look at the diagram showing part of the thermometer.



What is the temperature reading on the thermometer?

..... °C [1]

(ii) Tick (✓) which plan gives the most appropriate evidence for the investigation.

Measure the air temperature every day for one year.

☐

Measure the air temperature once a month for one year.

☐

Measure the air temperature once a week for one year.

☐

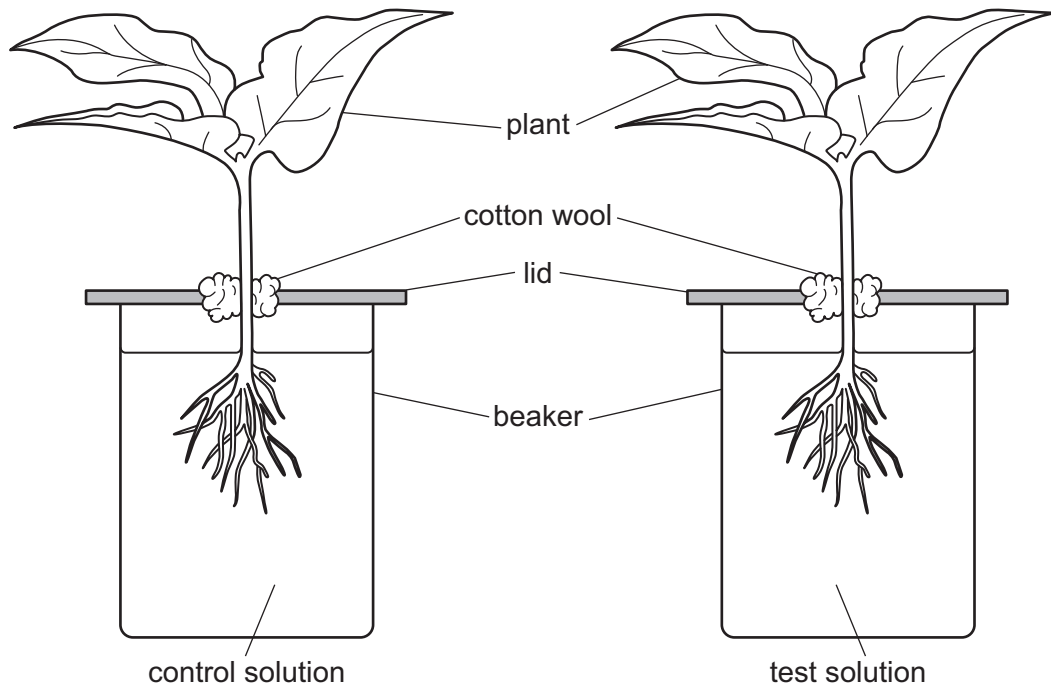
Measure the air temperature twice a day for one year.

☐

[1]

13 Pierre investigates how a lack of magnesium affects plants.

(a) The diagram shows the equipment he uses.



control solution contains all
the dissolved minerals a plant needs

test solution contains all
the dissolved minerals a plant needs
but **not** magnesium

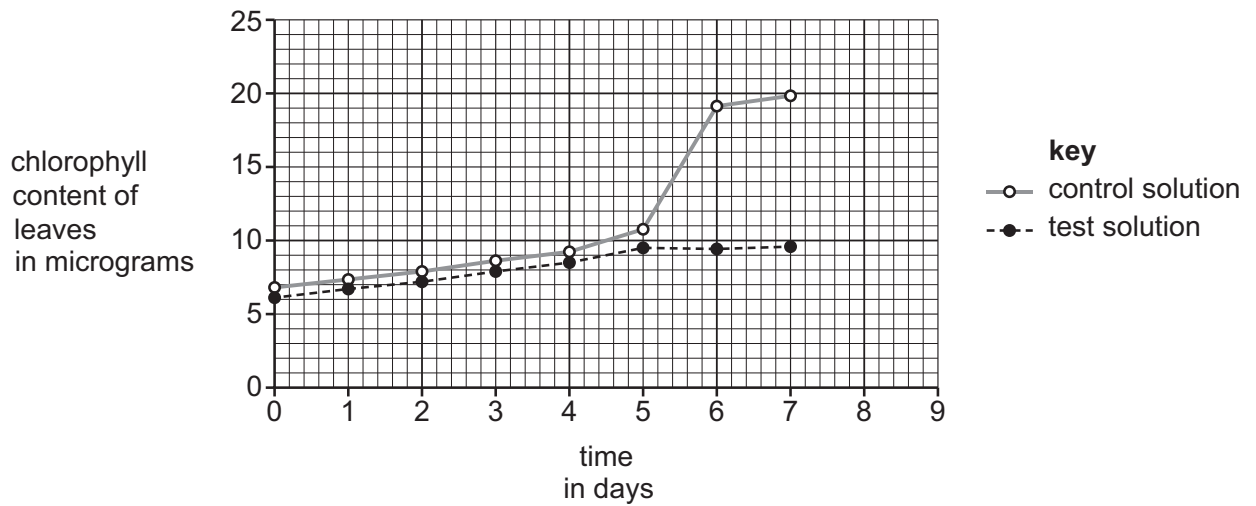
Pierre measures the chlorophyll content in the leaves of both plants every day.

(i) Pierre uses pure water to make the control solution and the test solution.

Explain why Pierre uses pure water and **not** water from a tap.

.....
..... [1]

(ii) Look at a graph of his results.



Write down **two** conclusions about how magnesium affects the chlorophyll content of leaves.

1

.....

2

.....

[2]

(b) Three solutions contain different percentages of minerals.

solution A

nitrates	2.0%
magnesium	1.0%

solution B

nitrates	6.0%
magnesium	0.2%

solution C

nitrates	4.0%
magnesium	1.2%

Which solution increases photosynthesis the most?

.....

Explain why.

.....

.....

[1]

14 Angelique makes an electrical circuit with a buzzer and a lamp.

(a) Draw the standard electrical symbol for a buzzer.

[1]

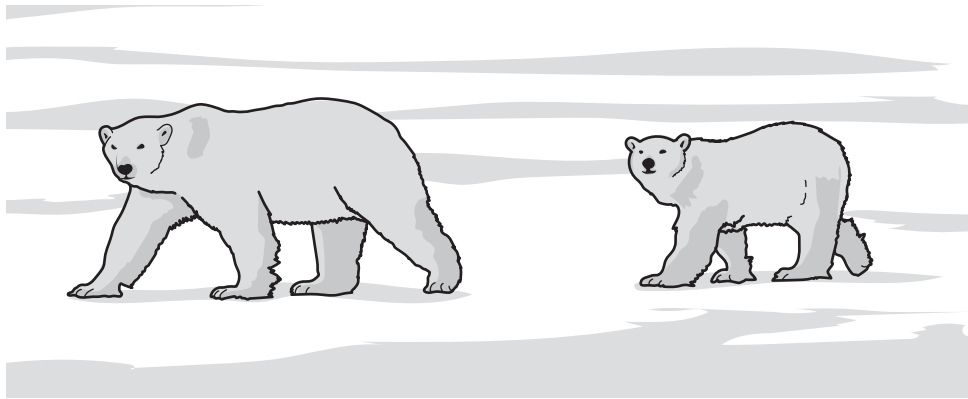
(b) She uses an electrical component to control the brightness of the lamp.

Draw the standard electrical symbol for this component.

[1]

15 Polar bears live in the Arctic.

Their main habitat is sea ice.



Polar bears use the sea ice when they are hunting for food.

This sea ice is melting.

Suggest how sea ice melting affects the polar bear population.

Give **two** reasons for your answer.

polar bear population

reason 1

.....

reason 2

.....

[3]

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The Periodic Table of Elements

Group																		
1	2	Key										3	4	5	6	7	8	
		1 H hydrogen 1																2 He helium 4
3 Li lithium 7	4 Be beryllium 9	atomic number atomic symbol name relative atomic mass											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24												13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —
lanthanoids																		
57 La lanthanum 139		58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175			
89 Ac actinium —		90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americum —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —			
actinoids																		



Science

Stage 9

Paper 2

2024

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

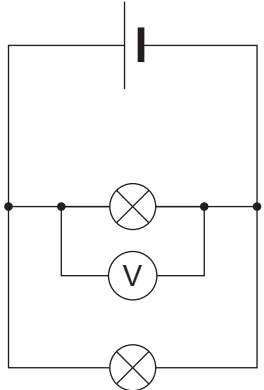
/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)(i)	photosynthesis	1	
1(a)(ii)	oxygen carbon dioxide	2	each correct answer = 1 mark Accept either order Accept O ₂ and CO ₂ Ignore water vapour
1(b)	(idea that) absorbs light (energy)	1	Ignore to help with photosynthesis

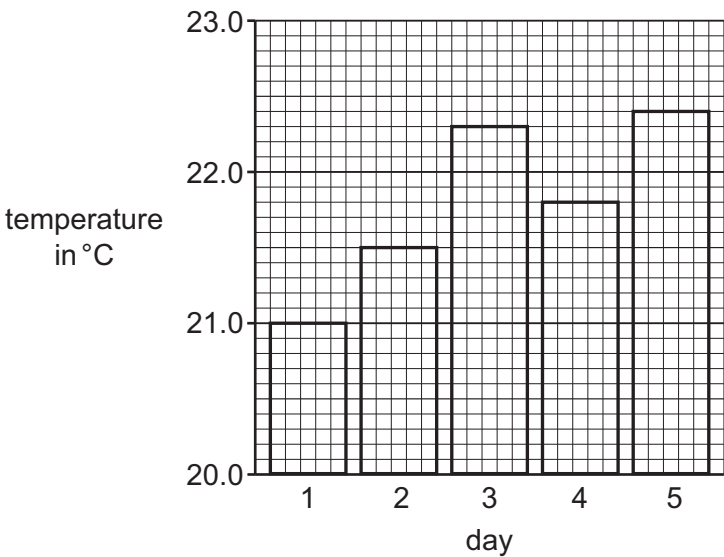
Question	Answer	Marks	Further Information
2(a)	(sodium atom) loses one electron (chlorine atom) gains one electron	2	each correct answer = 1 mark Accept sodium atom gains an electron and chlorine atom loses an electron or sodium atom loses electrons and chlorine atom gains electrons = 1 mark if no other mark is scored
2(b)	ions are attracted to each other (because) they are oppositely charged	2	each correct answer = 1 mark Ignore ionic bond Accept electrostatic attraction between ions = 2 marks Accept attraction between positive ions and negative ions = 2 marks

Question	Answer			Marks	Further Information
3	sentence	true	false	4	all eight correct = 4 marks six or seven correct = 3 marks four or five correct = 2 marks two or three correct = 1 mark one correct = 0 marks Note a tick in both boxes for a sentence means that sentence is incorrect Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
	Heat is the word used to measure how hot an object is.	☐	☒		
	Thermal energy is measured in °C.	☐	☒		
	When thermal energy is transferred more thermal energy is created.	☐	☒		
	Thermal energy always transfers from hotter objects to colder objects.	☒	☐		
	Heat dissipation happens when thermal energy transfers from a hotter region to a colder region.	☒	☐		
	A liquid is a good thermal energy conductor.	☐	☒		
	Convection happens in liquids and gases only.	☒	☐		
	Radiation needs particles to transfer thermal energy.	☐	☒		

Question	Answer	Marks	Further Information
4(a)	volcanic eruption or mountain building or earthquakes	1	Accept seismic activity Accept volcano or description of volcanic eruption
4(b)	convection (currents)	1	Accept movement of magma

Question	Answer	Marks	Further Information
5		2	voltmeter symbol = 1 mark Note ignore wire through voltmeter symbol if drawn in series voltmeter connected in parallel across one lamp = 1 mark Ignore junction of conductors

Question	Answer	Marks	Further Information
6(a)	copper carbonate + hydrochloric acid → copper chloride + water + carbon dioxide	2	correct reactants any order = 1 mark correct products any order = 1 mark
6(b)	filtration or filtering heat(ing) crystallise	3	each correct answer = 1 mark Accept using a funnel and filter paper Accept boiling Accept evaporate Accept form crystals
6(c)	goggles	1	Accept local name for safety goggles Ignore gloves or heat-proof mat or laboratory coat or fume cupboard or fume hood Do not accept face mask

Question	Answer	Marks	Further Information
7(a)	 temperature in °C day	2	graph with all five bars correct = 1 mark Accept $\pm \frac{1}{2}$ small square tolerance same thickness bars and same spacing between the bars = 1 mark Ignore any space between the first bar and the y-axis Ignore shading of bars
7(b)	any one from identifies anomalous results to take an average increases reliability	1	Do not accept more accurate

Question	Answer	Marks	Further Information
8	any three from Moon rock contains material that forms quickly at high temperatures Moon rock contains less iron than Earth rock (as collision blew away crust and mantle in debris) Moon rock has less water than Earth rock (as collision evaporated all of water) Moon rock and Earth rock are very similar in (chemical) composition if the Moon was created somewhere else, it would have a very different (chemical) content	3	each correct answer = 1 mark Accept Moon rock and Earth rock have the same (chemical) content

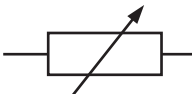
Question	Answer	Marks	Further Information
9(a)	D	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling on answer line takes precedence
9(b)	E	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling on answer line takes precedence
9(c)	DNA	1	Accept deoxyribonucleic acid

Question	Answer	Marks	Further Information
10	(density =) mass / volume or 150 / 100 (density =) 1.5 (unit) g / cm ³	3	Accept 1.5 g / cm ³ with or without working = 3 marks Accept 1.5 with or without working = 2 marks correct equation or substitution = 1 mark correct units = 1 mark

Question	Answer	Marks	Further Information
11(a)	no ticked (no mark) and the current is less or the resistance is more	1	if yes ticked = 0 marks if both boxes are ticked = 0 marks if neither box ticked accept no within the reason
11(b)	 (resistance =) voltage $\div$ current or $1.5 \div 4.5$ (resistance =) 0.33 (Ω)	2	Accept correct answer with or without working = 2 marks Accept equation or evidence of equation, e.g. correct substitution = 1 mark Accept $V \div I$ Accept 0.3 or 0.333 or 0.3333 etc. Ignore any unit given

Question	Answer	Marks	Further Information
12(a)	any two from more (scientific) evidence or data better understanding (of the evidence) models have improved carbon dioxide or pollutants (in the air) increased (more than expected)	2	each correct answer = 1 mark Ignore they have changed their mind
12(b)(i)	22.8 (°C)	1	
12(b)(ii)	Measure the air temperature twice a day for one year.	1	more than one answer ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence

Question	Answer	Marks	Further Information
13(a)(i)	(idea that) (pure water) does not contain impurities or (pure water) does not contain other minerals	1	Accept ora , e.g. tap water is impure Ignore universal solvent or dissolves minerals or it's not dirty
13(a)(ii)	any two from (idea that) magnesium has no effect on chlorophyll content between 0 and 5 days (idea that) after 5 days magnesium increases chlorophyll content (idea that) it takes 5 days to make chlorophyll after magnesium is introduced	2	each correct answer = 1 mark Accept Mg for magnesium Accept Mg for magnesium Accept Mg for magnesium
13(b)	C (no mark) and highest percentage of magnesium (for chlorophyll and sufficient nitrate for leaf growth)	1	Note if A or B = 0 marks

Question	Answer	Marks	Further Information
14(a)		1	Ignore wires from the buzzer
14(b)		1	

Question	Answer	Marks	Further Information
15	(polar bear population) (idea of) population decrease or extinction (reason) any two from (idea of) decrease in reproduction (idea of) less places to rest or sleep or give birth (idea of) less food or loss of hunting grounds or starvation	3	correct effect = 1 mark each correct reason = 1 mark Accept more cubs die or more bears die or polar bears now endangered Accept correct answers wherever seen

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